

LESSON 6.3

CONVERTING UNITS OF LENGTH



LESSON INTRODUCTION

MA.7.AR.3.3 Solve mathematical and real-world problems involving the conversion of units across different measurement systems.

LEARNING TARGETS

- I can convert length measurements across different measurement systems.

BENCHMARK COHERENCE

BUILDING ON

MA.6.AR.3.5 Solve mathematical and real-world problems involving ratios, rates, and unit rates, including comparisons, mixtures, ratios of lengths, and conversions within the same measurement system.

WORKING TOWARD

MA.8.AR.3 Extend understanding of proportional relationships to two-variable linear equations.

ESSENTIAL QUESTIONS

- How do you determine which unit of measure to place in the numerator and denominator of the conversion factor when solving a problem?
- How do you convert units of length between the U.S. customary and metric systems?

LESSON NARRATIVE

Students build on their working knowledge of conversions by applying those skills in contexts of measurement systems, focusing on units of length in this lesson. This lesson provides a conversation table for students as they explore and navigate a variety of applications for converting between inches, centimeters, and other common forms of measuring length in the U.S. customary and metric systems.

COMPONENT SUMMARY

6.3.1 WARM-UP (~5 MIN.)

Comparing inches and centimeters on a ruler

6.3.3 BRING IT TOGETHER (5 MIN.)

Comparing measuring systems

6.3.5 COLLABORATION (5 MIN.)

Exploring conversions using unit rates

6.3.7 YOUR TURN! (5 MIN.)

Additional conversion practice

6.3.2 ACTIVITY (10-15 MIN.)

Measuring objects using inches and centimeters to determine conversions

6.3.4 GUIDED PRACTICE (10 MIN.)

Using conversion tables

6.3.6 TRY IT (5 MIN.)

Practice converting lengths across systems

6.3.8 WRAP-UP (~5 MIN)

Evaluating student understanding and application

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- N/A

MATERIALS

- Measuring tools (ruler, tape, etc.)
- (Optional) Objects from home to measure
- Calculators

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may inadvertently apply the same conversion rate to different contexts. For example, in 6.3.6 Try It, students may multiply by 2.54 when converting from centimeters to inches (rather than dividing or multiplying by 0.394). Encourage students to include units in their conversion factors to help eliminate such errors.

THINGS TO CONSIDER

- Monitor for students using terms like “cancel out” to refer to the elimination of units. Use talk moves like “restate” to include mathematically correct vocabulary.
- Highlight student work that labels units correctly and build on errors as learning opportunities. The next few lessons all require precision in units and labeling.

LITERACY AND LANGUAGE ROUTINES

Contemplate, then Calculate

Building fluency includes decision-making in making a plan determining which method of conversion is most efficient. Students who first identify the method of conversion and verify for reasonableness will be less likely to make errors and perpetuate misconceptions on conversions. Try this routine to build fluency in 6.3.4 Guided Practice.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.3.1 Mathematical Fluency

Throughout the lesson, students continue to get practice with converting various units of measure, building upon Lessons 1 and 2. While the units of measure have changed, the process is very similar. As students use more precise vocabulary and understand the nuances of determining how to arrange the conversion factor, they should be working toward completing such tasks with accuracy and increased confidence.

DIFFERENTIATE INSTRUCTION

94.7% of the world’s population uses the metric system. Consider the population and lived experiences of your students in preparing and implementing this lesson. Does your classroom have many students and families that use the metric system at home or are more familiar with metric than U.S. systems of measurements? If so, consider incorporating student interest into the 6.3.2 Activity and examples and opportunities for students to discuss different systems and why they prefer one system over the other.

6.3.1 WARM-UP: A CLOSER LOOK AT RULERS

DIFFERENTIATE INSTRUCTION

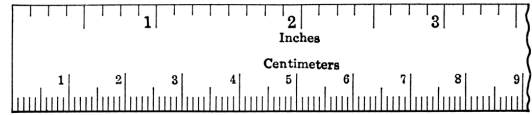
Consider sentence stems or explicit prompts for students to complete their three observations.

- “When comparing _____ to _____, I notice that _____.”
- “Inches are sectioned by _____ while centimeters are sectioned by _____.”
- Which is larger, one inch or one centimeter?



6.3.1 Warm-Up: A Closer Look at Rulers

A screenshot of a ruler that shows inches (in.) and centimeters (cm) is shown. The image is not to scale.



Look closely at the markings and labels on the ruler.

1. Write three observations about the markings and measurements on the ruler.

Answers will vary. The inches portion of the ruler is broken into 8 sections per inch. The centimeters portion of the ruler is segmented into 10 sections per centimeter. One-inch equals approximately 2.5 centimeters.

2. Compare observations with your partner. Add any additional observations your partner made to your list.



6.3.2 Activity: Measuring Objects to Determine Conversion Rates

1. With your partner, choose three classroom objects that can be measured with a standard ruler. Measure each object’s length in inches (in.) and in centimeters (cm). Use the measurements to complete the table, being as precise as possible.

Answers will vary.

Object	Length in Inches (in.)	Length in Centimeters (cm)
Pencil	7.5 in.	19 cm.
Sheet of Paper	11 in.	27.9 cm.
Shoe	11.7 in.	29.7 cm.

Similar to using exchange rates to convert currencies, you can use conversion rates to convert measurements.

To convert the length from inches (in.) to centimeters (cm), multiply the length in inches by a conversion rate. The result is the length in centimeters.

2. Use the two measurements for each object you measured to estimate a conversion rate for converting inches to centimeters.

Object 1	$7.5 \times r_1 = 19$	$r_1 \approx 2.54$
Object 2	$11 \times r_2 = 27.9$	$r_2 \approx 2.54$
Object 3	$11.7 \times r_3 = 29.7$	$r_3 \approx 2.54$

3. What is an estimate for the conversion rate of **inches to centimeters**?
 $r \approx 2.54$

To convert the length from centimeters to inches, multiply the length in centimeters (cm) by a conversion rate. The result is the length in inches (in.).

4. For each object, use the two measurements to estimate conversion rate for converting centimeters to inches.

Object 1	$19 \times r_4 = 7.5$	$r_4 \approx 0.394$
Object 2	$27.9 \times r_5 = 11$	$r_5 \approx 0.394$
Object 3	$29.7 \times r_6 = 11.7$	$r_6 \approx 0.394$

5. What is an estimate for the conversion rate of **centimeters to inches**?
 $r \approx 0.394$

6.3.2 ACTIVITY: MEASURING OBJECTS TO DETERMINE CONVERSION RATES

DIFFERENTIATE INSTRUCTION

If your classroom has only one shared measuring tool, consider having students create their own measuring unit to compare. For example, students could measure a nail or pencil eraser or other object using the shared ruler, then use that new form of unit in the table in substitute of centimeters or inches. For example, “This paper’s height is equal to 6.5 pencil erasers, which means it is 8.5 inches tall.”

QUESTIONING

Consider prompting groups who finish early to compare their values for r to the conversion table in 6.3.4 Guided Practice.

- What do you notice?
- What do you wonder?

TEACHER MOVES

Monitor for students who notice that dividing the values will produce the most precise value, or later, the conversion rate.

6.3.3 BRING IT TOGETHER: MEASUREMENT SYSTEMS



6.3.3 Bring It Together: Measurement Systems

Inches (in.) are a common unit for measuring length in the U.S. customary system, while centimeters (cm) are a common unit for measuring length in the metric system.

Did you know?



Most countries use the metric system of measurement. In fact, only three countries still use the customary system – the United States, Liberia, and Myanmar.

Conversion tables are a useful tool for converting one measurement system to another.

6.3.4 GUIDED PRACTICE: USING CONVERSION TABLES

ENRICHMENT

Compare and Connect

Position students as storytellers and experts of their lived experiences. Have students compare and connect their own experiences in addition to the purpose of different countries having different systems of measurement.

Use the conversion table and prompt students with the following questions:

- What forms of measurement have you or your family used?
- In your opinion, which system makes more sense to use? (Students may note the multiples of ten that make the metric system “easier.”)



6.3.4 Guided Practice: Using Conversion Tables

The tables below show the conversion rates for common units of length in the US Customary and Metric systems.

U.S. Customary System to Metric System

1 inch (in.)	= 25.4 millimeters (mm)
1 inch (in.)	= 2.54 centimeters (cm)
1 foot (ft)	= 30.48 centimeters (cm)
1 foot (ft)	= 0.305 meters (m)
1 yard (yd)	= 91.44 centimeters (cm)
1 yard (yd)	= 0.914 meters (m)
1 mile (mi)	= 1609.34 meters (m)
1 mile (mi)	= 1.61 kilometers (km)

Metric System to U.S. Customary System

1 millimeter (mm)	= 0.039 inches (in.)
1 centimeter (cm)	= 0.39 inches (in.)
1 centimeter (cm)	= 0.033 feet (ft)
1 meter (m)	= 39.37 inches (in.)
1 meter (m)	= 3.28 feet (ft)
1 meter (m)	= 1.094 yards (yd)
1 kilometer (km)	= 1093.613 yards (yd)
1 kilometer (km)	= 0.62 miles (mi)

Complete the statements below using the information from the conversion tables.

- To convert feet to centimeters, multiply the number of feet by 30.48 to find the equivalent number of centimeters.
- To convert centimeters to feet, multiply the number of centimeters by 0.033 to find the equivalent number of feet.
- To convert meters to yards, multiply the number of meters by 1.094 to find the equivalent number of yards.
- To convert yards to meters, multiply the number of yards by 0.914 to find the equivalent number of meters.

- Use the conversion rates to convert each measurement length.

Answers will vary depending on the conversion used.

- 30 miles $\approx$ 48.3 kilometers
 $30 \cancel{\text{mi}} \times \frac{1.61 \cancel{\text{km}}}{1 \cancel{\text{mi}}}$
- 65 kilometers $\approx$ 40.3 miles
 $65 \cancel{\text{km}} \times \frac{0.62 \cancel{\text{mi}}}{1 \cancel{\text{km}}}$
- 10 inches $\approx$ 254 millimeters
 $10 \cancel{\text{in}} \times \frac{25.4 \cancel{\text{mm}}}{1 \cancel{\text{in}}}$
- 450 millimeters $\approx$ 17.55 inches
 $450 \cancel{\text{mm}} \times \frac{0.039 \cancel{\text{in}}}{1 \cancel{\text{mm}}}$

6.3.4 GUIDED PRACTICE: USING CONVERSION TABLES (CONTINUED)

TEACHER MOVES

Consider having students complete questions 1-4 after the Compare and Connect routine above. Note the strategy of making a plan, similar to the upcoming routine Contemplate, then Calculate.

TEACHER MOVES

Contemplate, then Calculate

Similar to questions 1-4 above, consider intentionally requiring that students first identify their plan before putting pencil to paper. Thinking about the approach and comparing with a partner first will contribute to fewer mistakes and misconceptions.



6.3.5 Collaboration: Digging Deeper into Conversion Rates

The calculations below show how 5 feet (ft) can be converted to centimeters.

$$\frac{5 \cancel{\text{ft}}}{1} \times \frac{30.48 \cancel{\text{cm}}}{1 \cancel{\text{ft}}} = 152.4 \text{ cm}$$

Koji said the factor $\frac{30.48 \cancel{\text{cm}}}{1 \cancel{\text{ft}}}$ is equivalent to 1 because the numerator and denominator are equivalent values.

- With your partner, discuss what Koji used to say that $\frac{30.48 \cancel{\text{cm}}}{1 \cancel{\text{ft}}}$ is equivalent to 1.
He used the conversion table because $30.48 \text{ cm} \approx 1 \text{ ft}$.
- Explain how this shows 5 ft is equal to 152.4 cm.
Multiplying by 1 does not change the value.
- Eva noticed that in the calculation of the conversion, the unit feet (ft) was in the numerator and the denominator. How does this eliminate the units of feet in the calculation?
Similar to when the same value is in the numerator and denominator, the units become 1.
- Complete the statements below.
 When multiplying a measurement by a conversion factor, the value of the measurement
☐ does
☒ does not

 change because the conversion factor is equivalent to 1.
 The units
☒ do
☐ do not
 change because when the same units appear in the numerator and denominator, it is similar to having the same value in the numerator and denominator, so these units become 1.



Multiplicative Identity Property of One

$$a \times 1 = a \text{ OR } 1 \times a = a$$

6.3.5 COLLABORATION: DIGGING DEEPER INTO CONVERSION RATES

DIFFERENTIATE INSTRUCTION

The questions are intentionally scaffolded to position students to make connections between the method of conversion and the resulting units. Notice question 1 only asks students to discuss; this is in order to give students the opportunity to make sense of what they are seeing.

ENRICHMENT

Prompt students to make sense of what it means to “become 1” in their own words.

6.3.6 TRY IT: CONVERTING LENGTHS TO SOLVE PROBLEMS

TEACHER MOVES

Some students could multiply 16.8 and 0.394 to get 6.6192, while other students may notice that 16.8 could also be divided by 2.54 which is approximately 6.6142. Is one method preferred over the other? Why or why not?

Students might believe 6.6192 to be the answer instead of finding the difference.



6.3.6 Try It: Converting Lengths to Solve Problems

- Raphael measures the height of a plant each week using an inch (in.) ruler. On week 5, he cannot find the inch ruler, so he uses a centimeter (cm) ruler instead. His measurements are shown.

Plant Height

Week	1	2	3	4	5	6
Height	3.4 in.	4.2 in.	5.0 in.	5.8 in.	16.8 cm	7.4 in.

How much taller, in inches, is the plant in week 5 than in week 4?

$$16.8 \text{ cm} \times \frac{0.39 \text{ in.}}{1 \text{ cm}} = 6.552 \text{ in.}$$

-5.8

0.752 in.

6.3.7 YOUR TURN!

TEACHER MOVES

Prompt students to explain their reasoning for which units are preserved. Notice if students are using terms like “cancel” or “cross out” and prompt for increased precision in their explanations of why it is possible to eliminate units through the process of converting by multiplying a conversion factor.

QUESTIONING

For an extension project, consider having students compare and contrast travel efficiency by comparing like miles per gallon to kilometers per charge.



6.3.7 Your Turn!

- Use the conversion rates to convert between the lengths below. Round to the nearest thousandth when necessary.

Answers will vary depending on the conversion used.

- 25 miles $\approx$ 40.25 kilometers
- 25 centimeters $\approx$ 0.825 feet
- 3 yards $\approx$ 2.742 meters
- 470 millimeters $\approx$ 18.33 inches

- Maya visits her friend in Europe and rides in an electric car. The electric vehicle can travel 200 kilometers (km) on one charge. If their destination is 130 miles (mi) away, will they be able to make it there without having to stop to charge the car? Explain or show work to justify your answer.

$$200 \text{ km} \times \frac{1 \text{ mi}}{1.61 \text{ km}} \approx 124.2 \text{ mi.}$$

*Answers will vary depending on the conversion used and the place value to which it is rounded.
Sample answer: They will not make the 130 miles trip on one charge. One charge will take them approximately 124 miles which is less than 130 miles.*



6.3.8 Wrap-Up: Ticket Out the Door

Darnell and Jada both walk every day for exercise. Darnell averages 6.8 kilometers (km) per day while Jada averages 4.5 miles (mi) each day. Darnell claims he walks farther than Jada, but Jada disagrees.

- Without performing any calculations, which student do you think is correct? Explain your reasoning.

Answers will vary.

Jada is correct. In Guided Practice #5 it showed that every 30 miles is equal to approximately 49 kilometers. This difference is bigger than the difference of the actual amount Darnell walks vs the amount Jada walks each day.

- Use unit conversions to determine who walks farther each day.

$$6.8 \text{ km} \times \frac{1 \text{ mi}}{1.61 \text{ km}} = 4.22 \text{ mi.}$$

Jada walks farther each day.

6.3.8 WRAP-UP: TICKET OUT THE DOOR

TEACHER MOVES

Consider prompting students to use at least one glossary or vocabulary term or an example from the lesson in their explanation to question 1. See the sample response provided in the key for one connection students could make.